

Analysis Plan for HAART Outcomes by Drug Use

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Introduction

This project uses data from the Multicenter AIDS Cohort Study, an ongoing prospective cohort that studies HIV-1 infection among homosexual and bisexual men in four major US cities. This report analyzed a subset of 463 MACS participants with viral load (log10 HIV copies/mL blood), CD4+ T cell count, and SF-36 aggregate physical and mental quality-of-life measured at baseline (pre-HAART initiation) and two years after starting HAART. Baseline information on hard drug use, defined as injection drugs or illicit heroin/opiate use, was available along with and other relevant covariates including age, BMI, smoking status, education, and race/ethnicity.

The primary question of interest was whether individuals who reported hard drug use at baseline experienced different treatment outcomes two years after starting HAART compared to those who did not after adjusting for baseline covariates. The outcomes were viral load (log10 HIV copies/mL blood), CD4+ T cell count, and SF-36 aggregate physical and mental quality-of-life. The covariates were age, BMI, smoking status, education, and race/ethnicity. The secondary analysis will descriptively assess whether the difference or lack of difference in treatment response between hard drug users and non hard drug users after adjusting for baseline covariates can be explained by year-2 HAART adherence. These two questions were both assessed from a frequentist and a Bayesian framework.

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Methods

All analyses were conducted in R v4.5.2 (2025-10-31 ucrt). BMI values coded as -1 or 999 and two implausible values over 250 were set to missing. Categorical covariates were collapsed as follows to ensure adequate sample sizes by category: smoking status (current vs former/non smoker), education status (no college degree vs college degree and above), race/ethnicity (white non-Hispanic vs other), and adherence (95% and above vs <95%). Viral load was converted to the log10 scale. Participants were removed if they were missing baseline after removing any individuals with incomplete baseline viral load, CD4+ T cell count, physical and mental quality of life scores, age, BMI, smoking status, education, or race/ethnicity. Participants were also removed if they were missing year two outcomes or year two adherence. The final sample size was 463 individuals.

For each outcome at year 2, linear regression was used to identify the association with hard drug use at baseline, adjusting for age, BMI, smoking status, education, and race/ethnicity. Frequentist models reported point estimates, 95% confidence intervals (CI), and unadjusted p-values. Bayesian models were implemented using STAN with vaguely informative priors as prior knowledge is limited: $N(0,1000)$ for regression coefficients and half-normal(0,1000) for residual SD. The intercept was not centered due to the large variance and model fit was diagnosed with R-hat and ESS. Bayesian models reported posterior means, 95% highest posterior density intervals (HPDI), and posterior probabilities. Posterior probabilities were calculated from clinically meaningful differences for each outcome (two-sided), defined as follows: 0.5 log10 (viral load), 50 cells (CD4+ T cell count), and 2 points (quality of life). These models were additionally fit including year 2 adherence and descriptively compared

to the original models.

Because only one participant in the hard-drug-use group had adherence $<95\%$, we conducted a sensitivity analysis using an alternative adherence grouping ($<100\%$ vs. 100%) to ensure that sparse data did not distort the adherence-adjusted estimates. This alternative definition increased the number of individuals classified as having suboptimal adherence from 1 to 18 within the hard-drug-use group, improving model stability. The Bayesian posterior means, HPDIs, and posterior probabilities under this alternative specification were nearly identical to those obtained using the investigator-specified $<95\%$ cutoff, and the corresponding frequentist estimates showed the same pattern. Across all outcomes, the direction, magnitude, and uncertainty of the adherence-adjusted associations changed only trivially, and the substantive interpretation remained unchanged. These findings indicate that the adherence definition does not materially influence the estimated relationship between hard drug use and year-2 clinical outcomes, and we therefore retained the investigator-requested adherence grouping for consistency with the study protocol.

Results

Table 1 summarizes key variables in the final sample of 463 at baseline across baseline hard drug use. *Table 2* summarizes key variables at year 2 across baseline hard drug use. *Figure 1* visualizes the distributions of year 2 outcomes by hard drug status.

The modeling outcomes are summarized in *Table 3*. The frequentist and Bayesian outcomes agree well, so only Bayesian outcomes will be stated in the text. For viral load, the posterior mean for the association with baseline hard drug use was -0.058 with a 95% HPDI of (-0.446,

0.342), indicating substantial posterior uncertainty and little evidence of a clinically meaningful effect, defined as a 0.5 log10 change. This is supported by the posterior probability of 0.016, which suggests that there was little support for a harmful association. For CD4+ T cell count, the posterior mean was -163.5 cells (95% HPDI: (-225.5, -100.7)). The posterior probability that the effect was negative was 1.00, strongly indicating a reduction in CD4 T cell count among individuals who reported baseline hard drug use. This reduction was clinically meaningful, as both sides of the 95% HPDI are greater than 50 cells. Differences in quality of life scores are defined as clinically meaningful when there is a change of 2 points. For physical quality of life, the posterior mean was -3.34 with a 95% HPDI of (-6.02, -0.64). Along with the posterior probability of 0.836, there is moderate evidence of reduced physical quality of life among those who reported hard drug use. In contrast, for mental quality of life, the posterior mean (-0.29, 95% HPDI: (-3.87, 3.05)) and posterior probability (0.265) do not provide strong evidence of any association.

Adjustments for adherence produced minimal changes in the Bayesian estimates for most outcomes, and the posterior distributions remained aligned with the frequentist results (*Table 4*). For viral load, the adherence-adjusted posterior mean was attenuated by 77% from -0.058 to -0.013 (95% HPDI: -0.416, 0.386) and the posterior probability remained low (0.014). These results maintain the conclusion that there is little support for any association between hard drug use and viral load regardless of the adherence adjustment. For CD4+ T cell count, the adherence-adjusted posterior mean differed by only -3.8% and the posterior probability remained 1.00. This continues to support the conclusion that there was a reduction in CD4+ T cell count at year 2 among baseline hard drug users and that adherence does not explain

the observed association. For physical quality of life, the posterior mean had a 6% change from the unadjusted model, and the posterior probability only modestly increased to 0.866 from 0.836, continuing to provide evidence for lower second year physical quality of life scores among baseline hard drug users. However, the small shifts do not suggest that adherence accounts for much of this association. For mental quality of life, the adherence-adjusted posterior mean was attenuated by 81% relative to the unadjusted model from -0.29 to -0.52 (95% HPDI: (-4.15, 2.78)). However, the posterior probability remained low (0.275), indicating little support for an association between hard drug use and mental quality of life regardless of the adherence adjustment.

Figure 2 visually shows the agreement between the frequentist and Bayesian approaches with and without adherence included in the model.

Discussion

TODO - edit names in tables and reorder groupings

Appendix

Tables and Figures

Table 1: Baseline Characteristics by Baseline Hard Drug Use

	No (N=427)	Yes (N=36)	Overall (N=463)
log10VLOAD_0			
Mean (SD)	4.53 (0.930)	4.58 (0.871)	4.53 (0.925)
Median [Min, Max]	4.52 [0.237, 8.28]	4.47 [2.87, 6.40]	4.51 [0.237, 8.28]
LEU3N_0			
Mean (SD)	374 (202)	360 (201)	373 (202)
Median [Min, Max]	360 [10.9, 1220]	397 [10.9, 650]	360 [10.9, 1220]
AGG_PHYS_0			
Mean (SD)	51.4 (9.00)	49.2 (7.05)	51.2 (8.88)
Median [Min, Max]	53.7 [19.2, 69.0]	47.4 [31.4, 62.9]	53.5 [19.2, 69.0]
AGG_MENT_0			
Mean (SD)	45.2 (13.8)	42.0 (11.6)	44.9 (13.6)
Median [Min, Max]	49.5 [7.23, 66.0]	44.4 [22.5, 59.6]	49.1 [7.23, 66.0]
age_0			
Mean (SD)	43.1 (8.68)	43.9 (9.52)	43.2 (8.74)
Median [Min, Max]	43.0 [20.0, 73.0]	45.5 [29.0, 61.0]	43.0 [20.0, 73.0]
BMI_0			
Mean (SD)	25.3 (4.39)	23.6 (3.45)	25.2 (4.34)
Median [Min, Max]	24.8 [16.5, 45.3]	23.3 [18.0, 31.2]	24.6 [16.5, 45.3]
smoke_group_0			
Current Smoker	149 (34.9%)	27 (75.0%)	176 (38.0%)
Non or Former Smoker	278 (65.1%)	9 (25.0%)	287 (62.0%)
educ_group_0			
College Degree and above	191 (44.7%)	10 (27.8%)	201 (43.4%)
No College Degree	236 (55.3%)	26 (72.2%)	262 (56.6%)
race_group_0			
Other	150 (35.1%)	17 (47.2%)	167 (36.1%)
White nonHispanic	277 (64.9%)	19 (52.8%)	296 (63.9%)

Table 2: Year 2 Characteristics by Baseline Hard Drug Use

	No (N=427)	Yes (N=36)	Overall (N=463)
log10VLOAD_2			
Mean (SD)	1.79 (1.21)	1.85 (1.45)	1.79 (1.23)
Median [Min, Max]	1.49 [-0.608, 5.85]	1.64 [-0.131, 5.63]	1.49 [-0.608, 5.85]
LEU3N_2			
Mean (SD)	556 (264)	371 (249)	542 (267)
Median [Min, Max]	536 [39.5, 1730]	357 [60.0, 971]	516 [39.5, 1730]
AGG_PHYS_2			
Mean (SD)	50.0 (9.93)	44.4 (12.0)	49.6 (10.2)
Median [Min, Max]	53.3 [14.8, 69.1]	44.8 [18.2, 63.9]	53.2 [14.8, 69.1]
AGG_MENT_2			
Mean (SD)	47.6 (11.8)	45.9 (14.1)	47.4 (12.0)
Median [Min, Max]	51.2 [10.5, 66.7]	49.2 [21.3, 65.3]	51.2 [10.5, 66.7]
adh_group_2			
<95%	45 (10.5%)	1 (2.8%)	46 (9.9%)
95% and Above	382 (89.5%)	35 (97.2%)	417 (90.1%)

Table 3: Comparison of Frequentist and Bayesian Estimates for Associations Between Hard Drug Use and Clinical Outcomes

Outcome	Frequentist			Bayesian		
	Point Estimate	95% CI	Unadjusted P-Value	Posterior Mean	95% HPDI	Posterior Probability
Viral Load (log10 HIV copies/mL blood)	-0.059	(-0.456, 0.337)	0.770	-0.058	(-0.446, 0.342)	0.016
CD4+ T Cell Count	-163.880	(-226.180, -101.580)	0.000	-163.520	(-225.483, -100.741)	1.000
Physical Quality of Life (SF-36 Aggregate)	-3.336	(-6.055, -0.617)	0.017	-3.344	(-6.016, -0.627)	0.836
Mental Quality of Life (SF-36 Aggregate)	-0.289	(-3.743, 3.165)	0.870	-0.290	(-3.873, 3.051)	0.265

Comparison of Frequentist and Bayesian regression estimates for the association between baseline hard drug use and year-2 outcomes. All models adjust for baseline outcome values, age, BMI, smoking status, education, and race. Bayesian models use weakly informative Normal(0, 1000) priors for regression coefficients and half-Normal(0, 1000) priors for residual SD.

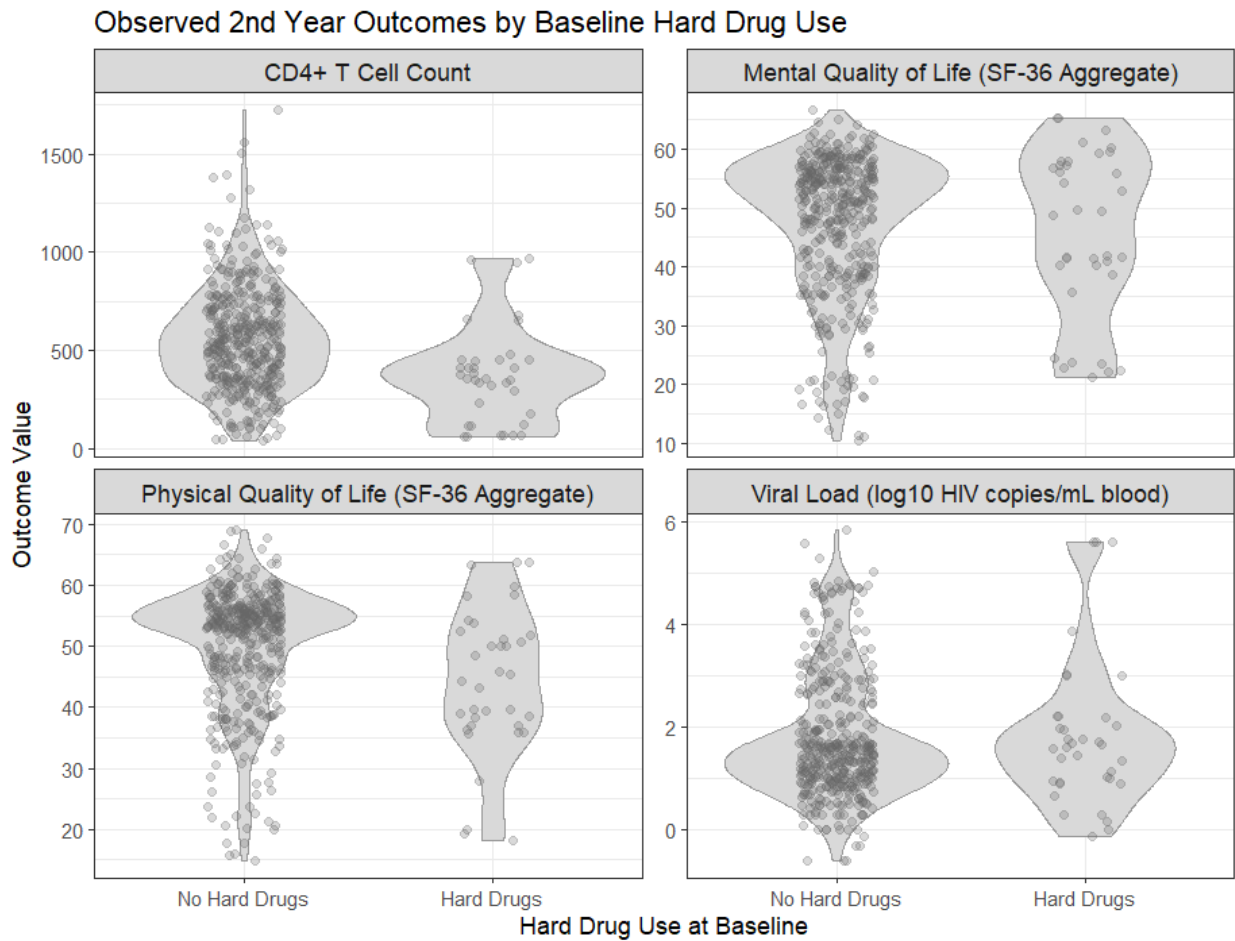


Figure 1: Year 2 observed outcomes by hard drug use at baseline.

Table 4: Impact of Adherence Adjustment on Frequentist and Bayesian Estimates for Associations Between Hard Drug Use and Clinical Outcomes

Outcome	Frequentist				Bayesian			
	Point Estimate	Change in PE (%)	95% CI	Unadjusted P-Value	Posterior Mean	Change in PM (%)	95% HPDI	Posterior Probability
Viral Load (log10 HIV copies/mL blood)	-0.053	9.725	(-0.451, 0.344)	0.792	-0.054	7.222	(-0.452, 0.343)	0.018
CD4+ T Cell Count	-162.218	1.014	(-224.558, -99.879)	0.000	-162.752	0.470	(-223.218, -98.692)	1.000
Physical Quality of Life (SF-36 Aggregate)	-3.221	3.460	(-5.935, -0.506)	0.020	-3.225	3.552	(-5.957, -0.513)	0.808
Mental Quality of Life (SF-36 Aggregate)	-0.506	-75.428	(-3.944, 2.932)	0.773	-0.513	-77.246	(-3.968, 2.942)	0.274

Frequentist and Bayesian regression estimates for the association between baseline hard drug use and year-2 outcomes after adjusting for adherence. All models also adjust for baseline outcome values, age, BMI, smoking status, education, and race. Percent change reflects the relative difference between adherence-adjusted and non-adjusted models, calculated for both the frequentist point estimate and the Bayesian posterior mean. Bayesian models use weakly informative Normal(0, 1000) priors for regression coefficients and half-Normal(0, 1000) priors for residual SD.

Directories Code Directory: <https://github.com/eweible/BIOS6624.git>

File name: P1/Code/P1_Report.Rmd

Data Directory: C:/Users/weiblee/OneDrive/Documents/Anschutz/Spring 2026/6624_Advanced/Weible_Advanced/P1/DataRaw

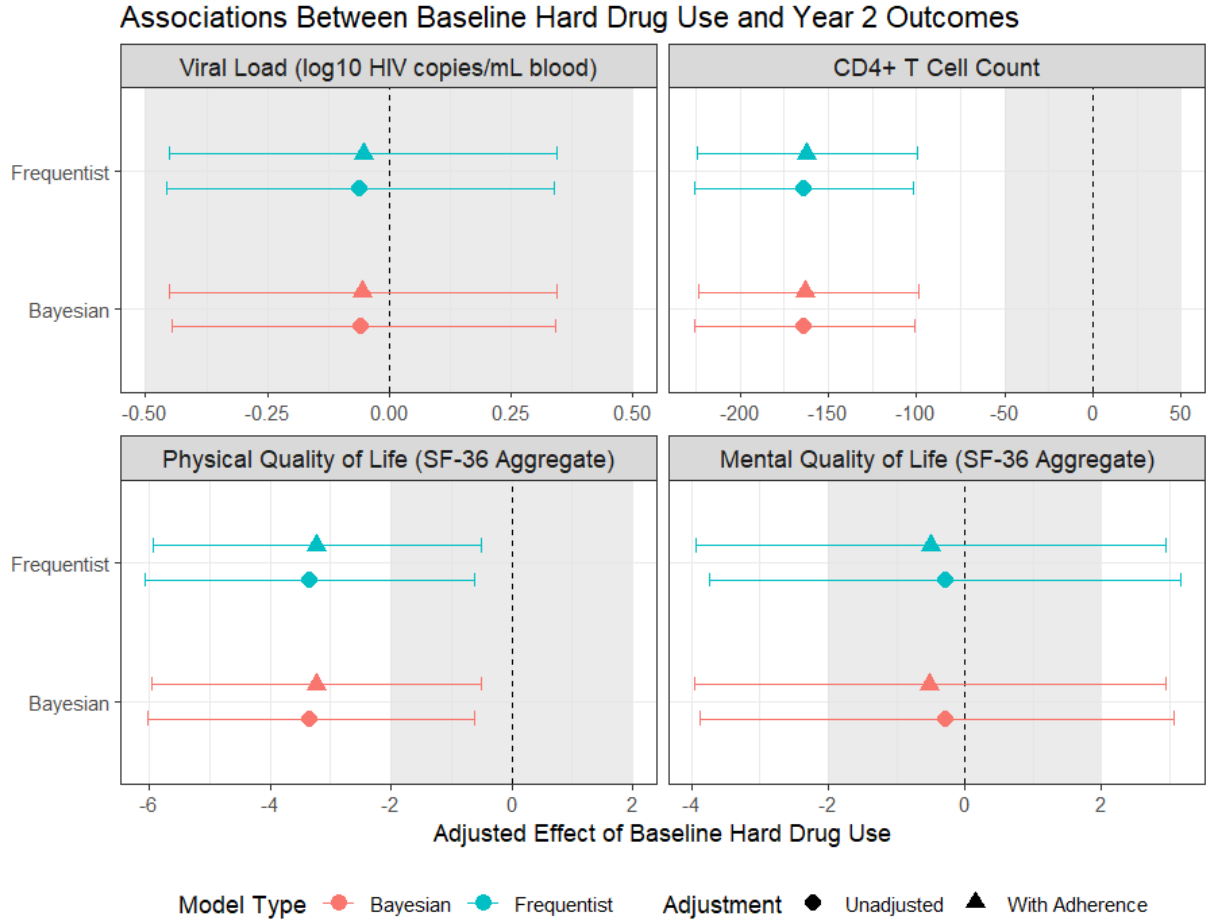


Figure 2: Adjusted associations between baseline hard drug use and 2nd year clinical outcomes estimated using Frequentist and Bayesian regression models, shown separately for models with and without adherence. All models adjust for baseline outcome values, age, BMI, smoking status, education, and race. Bayesian models use weakly informative Normal(0, 1000) priors for regression coefficients and half-Normal(0, 1000) priors for residual standard deviations. Points represent the estimated effect of baseline hard drug use, with error bars showing 95% confidence intervals (Frequentist) or 95% highest posterior density intervals (Bayesian). The grey shaded region in each panel denotes the range corresponding to the two-sided clinically meaningful difference for that outcome (0.5 log10 for viral load, 50 cells for CD4 count, and 2 points for quality of life scales), providing a visual benchmark for interpreting the magnitude of estimated effects.